E-COMMERCE SALES DATA ANALYSIS

Internship Project Report

The Developers Arena

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Technology: Python, Pandas, Matplotlib

Project Type: Data Analysis

This report presents the analysis, visualization, findings, and conclusions obtained from the provided e-commerce sales dataset.

1. Introduction

E-commerce generates a large amount of sales data every day. Analyzing this data helps businesses understand sales performance, product performance, regional trends, and purchasing patterns. This project analyzes an e-commerce sales dataset using Python. Pandas is used for data loading, cleaning, exploration, and analysis, while Matplotlib is used for visualization.

2. Project Objectives

The main objective is to analyze e-commerce sales data and extract meaningful business insights.

- Analyze overall sales performance.
- Identify the best-performing products.
- Compare sales across different regions.
- Calculate total quantity sold and average transaction value.
- Find the highest-value transaction.
- Identify sales trends over time.
- Present findings using clear charts and graphs.

3. Dataset Description

The supplied dataset contains **100 sales records** and **7 columns**. It includes 5 products and 4 regions.

Column	Description
Date	Date of the transaction
Product	Product sold
Quantity	Number of units sold
Price	Price of the product
Customer_ID	Unique customer identifier
Region	Sales region
Total_Sales	Total value of the transaction

4. Tools and Technologies

Python, Pandas, Matplotlib, CSV, VS Code, Git, and GitHub were used or planned for the project workflow.

5. Data Cleaning

The dataset was checked for missing values and duplicate records. The analysis found **0 missing values** and **0 duplicate rows**. The Date column was converted to datetime format for time-based analysis.

6. Exploratory Data Analysis

The dataset was explored using Pandas functions such as head(), shape, info(), describe(), isnull(), and duplicated(). These checks helped understand the structure, data types, statistical characteristics, and data quality before the main analysis.

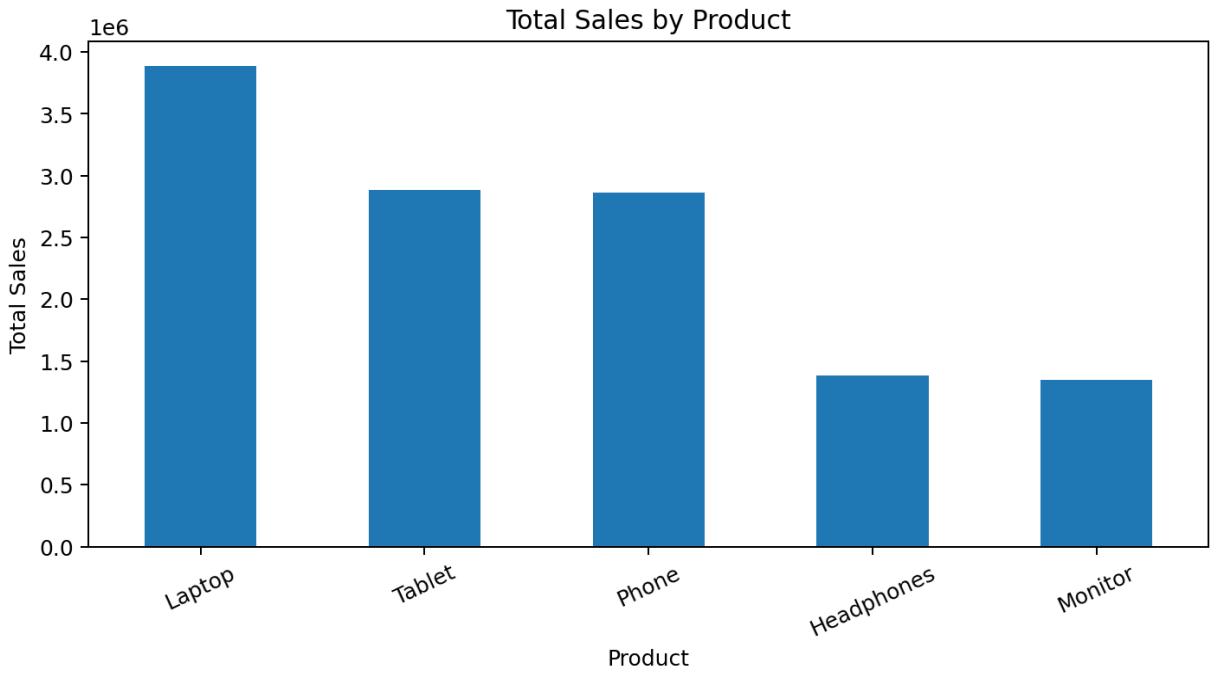
7. Overall Sales Analysis

The dataset contains **100** transactions. Total recorded sales are ■**12,365,048.00**. A total of **478** units were sold, with an average transaction value of approximately ■**123,650.48**.

Metric	Value
Number of transactions	100
Total sales	■12,365,048.00
Total quantity sold	478
Average transaction value	■123,650.48
Highest transaction	■373,932.00

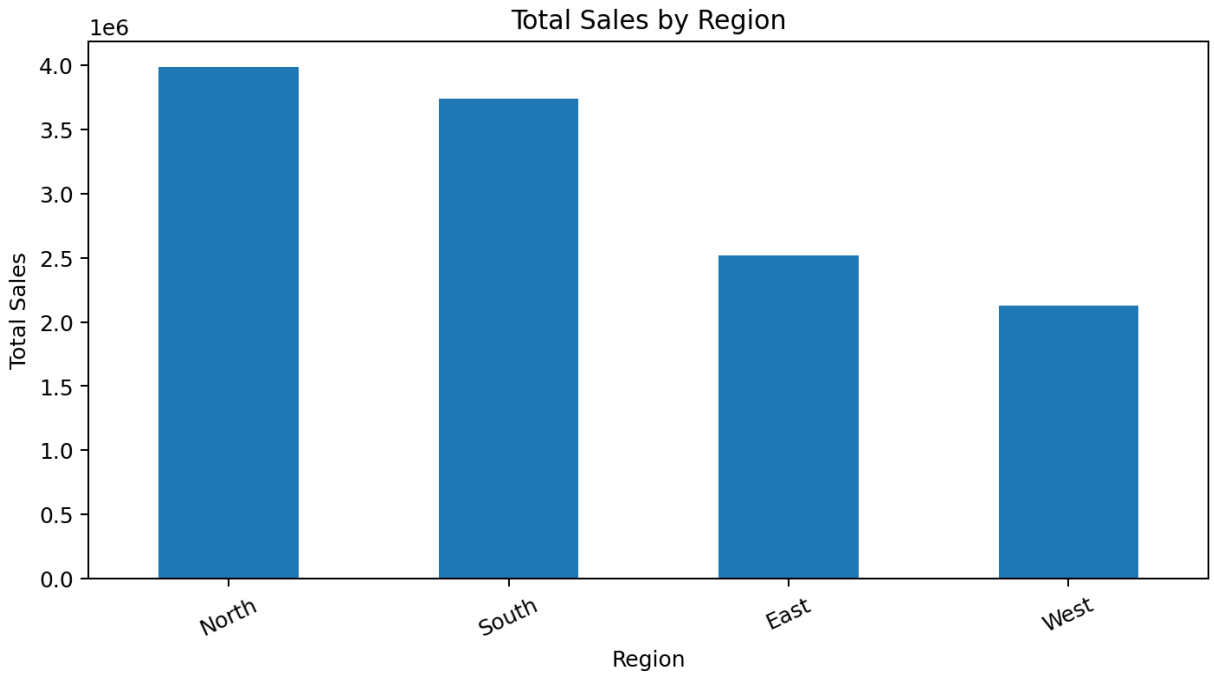
8. Product Analysis

Laptop generated the highest total sales, with revenue of ■**3,889,210.00**. Product-wise sales were calculated with Pandas groupby() and sorted to identify the strongest-performing products.



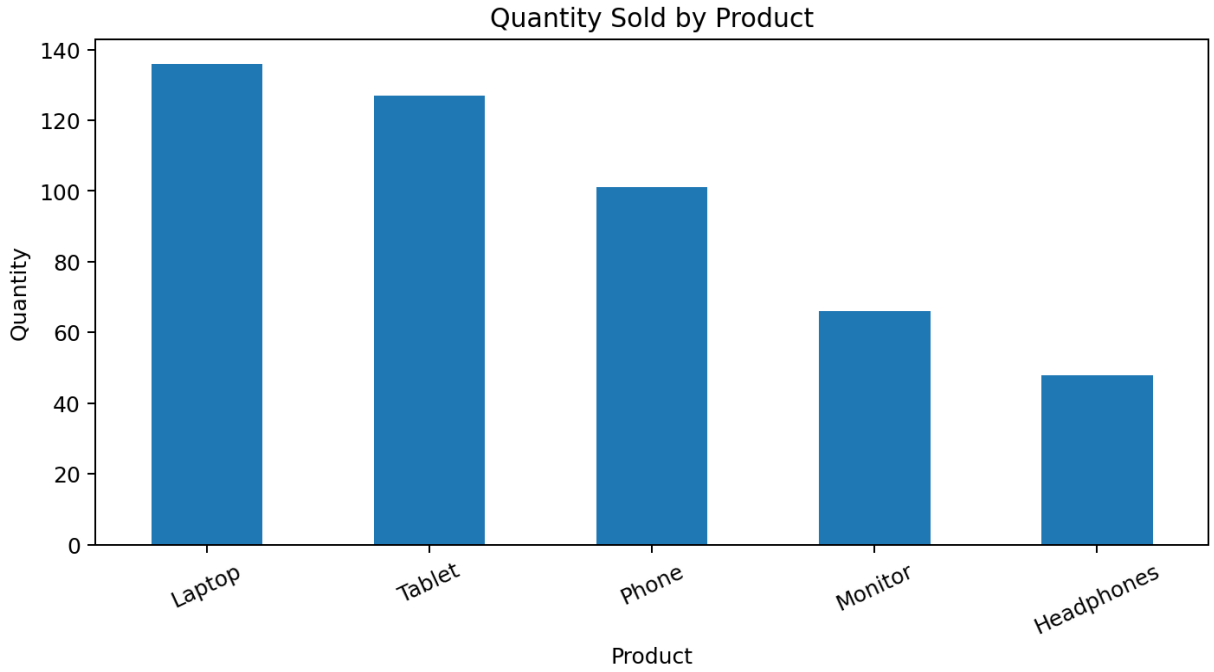
9. Regional Analysis

North generated the highest regional sales, with revenue of ■**3,983,635.00**.



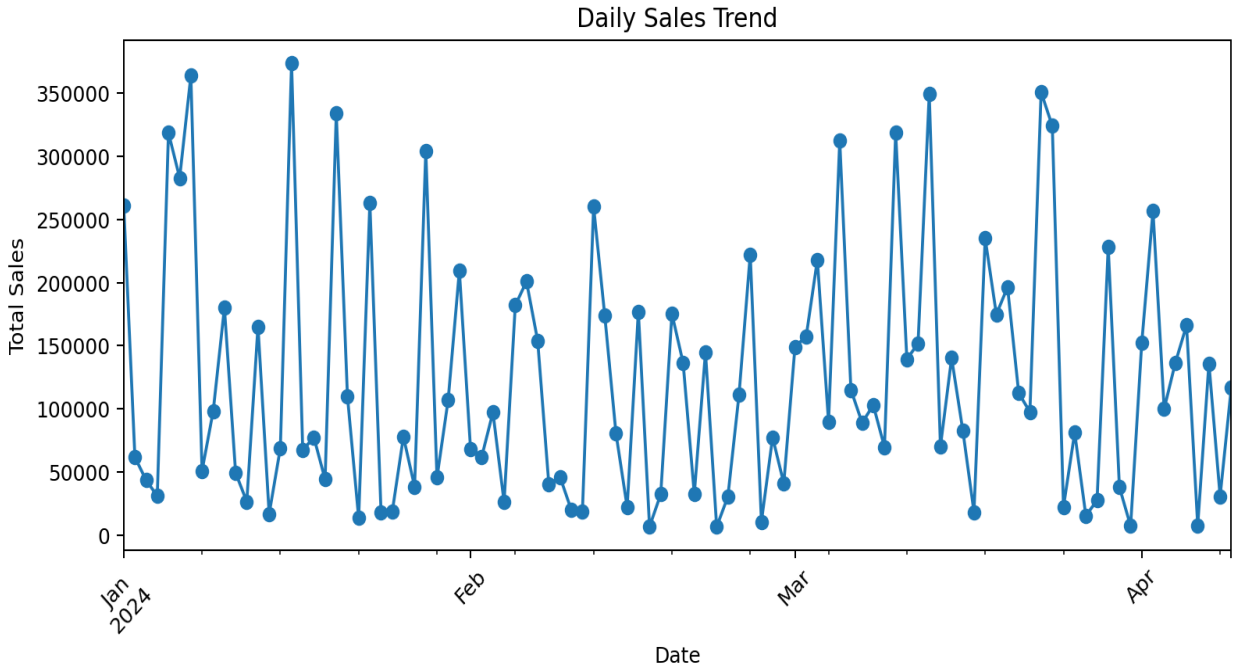
10. Quantity Analysis

Laptop had the highest total quantity sold, with **136** units.



11. Sales Trend Analysis

Daily sales were grouped by date and displayed as a line chart. This helps visualize changes and fluctuations in sales over the period covered by the dataset.



12. Key Insights

- The dataset contains 100 sales transactions across 5 products and 4 regions.
- Laptop is the highest-revenue product.
- North is the highest-performing region by total sales.
- Laptop has the highest number of units sold.
- No missing values or duplicate rows were found.
- Product-wise and region-wise aggregation makes sales performance easier to compare.

13. Conclusion

This project provided practical experience in performing data analysis using Python. The e-commerce sales dataset was loaded, cleaned, explored, analyzed, and visualized using Pandas and Matplotlib. The analysis identified product-wise and regional sales performance and demonstrated how visualizations can make patterns and trends easier to understand.

14. Future Scope

- Build an interactive dashboard using Power BI.
- Analyze larger real-world datasets.
- Perform customer segmentation.
- Add monthly and yearly sales comparisons.

- Predict future sales using machine learning.
- Analyze customer purchasing behavior.

15. Project Structure

```
Ecommerce-Sales-Analysis/
■■■ data/sales_data.csv
■■■ visualizations/
■■■ report/E-commerce_Sales_Analysis_Report.pdf
■■■ main.py
■■■ README.md
■■■ requirements.txt
```